Caution Is Not Discrimination: Measuring Inferential Overreach in LLM-Assisted Digital Forensics

Anonymous

Abstract

A language model can cite genuine digital evidence and still make a claim that evidence does not license. Given a browser history row it reports that a user deliberately visited a page, though a URL and a timestamp cannot separate a click from a redirect. We call this *inferential overreach* and study it on the transition from an observed trace to an activity-level proposition in the forensic hierarchy of propositions. We separate two requirements that evidential sufficiency collapses: evidence volume against evidence *kind*, and trace *presence* against the *relation* a trace stands in to the record at issue. The second holds the target fixed and measures relational discrimination, not causal provenance. It is anchored on file execution, where the relation is executable identity, and stress-tested on browser history, where it must be inferred from timing. Across five models, two corpora and two artifact families, overreach under plain prompting runs from 39% to 92% of the browser conditions whose evidence licenses nothing, and the two contrasts show different patterns across models. Naming the forensic criterion does not repair this. On three of five models it drives retention on the single-record condition to zero, cutting unsupported claims by removing supported ones, not by discriminating. Naming the relation recovers supported-claim retention relative to the generic criterion on every model, and on the main corpus separates the near decoy at least as often. A rule with access only to trace type cannot express that relation in either family, so the limit is structural, and whether models cross it appears to depend on whether the relation is recorded or inferred.

Keywords: digital forensics, large language models, hierarchy of propositions, activity level, evidential sufficiency, evaluation methodology

1. Introduction

Consider a single row recovered from a Chrome history database:

```
2015-03-23 15:55:34
http://www.icloud.com/icloudcontrolpanel
Title:  iCloud
```

Asked what this shows, a language model will answer that the user accessed the iCloud control panel. The artifact is real, the timestamp is real, and the model has invented nothing. Yet the record above is a URL, a time and a title, and at that level a browser writes the same thing for a redirect hop or an automatically loaded subframe as it does for a click. The record establishes that the URL was loaded. It does not establish that a person loaded it.

We call this *inferential overreach*: a claim correctly attributed to real evidence but pitched at a level of proposition that evidence does not license. The failure need not involve a fabricated artifact or a wrong citation: it can arise from an unsupported activity-level inference drawn from genuine traces. Faithfulness measures that decompose an answer and test entailment against the retrieved context [18, 19] may well mark the claim unsupported, but they score one axis and carry no representation of what the jump consists of or of what evidence would close it. The question we put is not *Evidence* $\models$ *Claim* but *which proposition level does this evidence permit?*

Forensic science has settled vocabulary for exactly that distinction. Cook et al. [1] separate propositions about the source of a trace, the activity that produced it, and the offence. The levels differ, they observe, not in confidence but in what must be known to address them, and ENFSI's guideline for evaluative reporting [3] requires a report to state the level at which its propositions sit. That hierarchy governs what a human examiner may write. To our knowledge it has not been used as an evaluation constraint on what an automated generator is entitled to state.

The gap is therefore not whether a model is grounded in evidence but whether the evidence it cites licenses a proposition at the level it asserts, and groundedness and abstention measures, as they are commonly formulated, do not validate that crossing. This paper supplies a protocol that does: it builds evidence conditions in which licensing is controlled by construction, and reads a generator's verdicts against them. Figure 1 sets out the design. Its first move is to separate two requirements that "sufficient evidence" runs together.

Volume against type: one record can settle a question that five cannot. *Presence against record relation:* a trace of the right type may stand in no relation to the record under examination. Section 3.3 builds a condition for each. Against both we compare generic evidential caution, naming the forensic prerequisite, and asking that prerequisite as a separate question. These are instruments for locating the boundaries, not proposals: we offer no architecture and no prompt as a contribution, and one

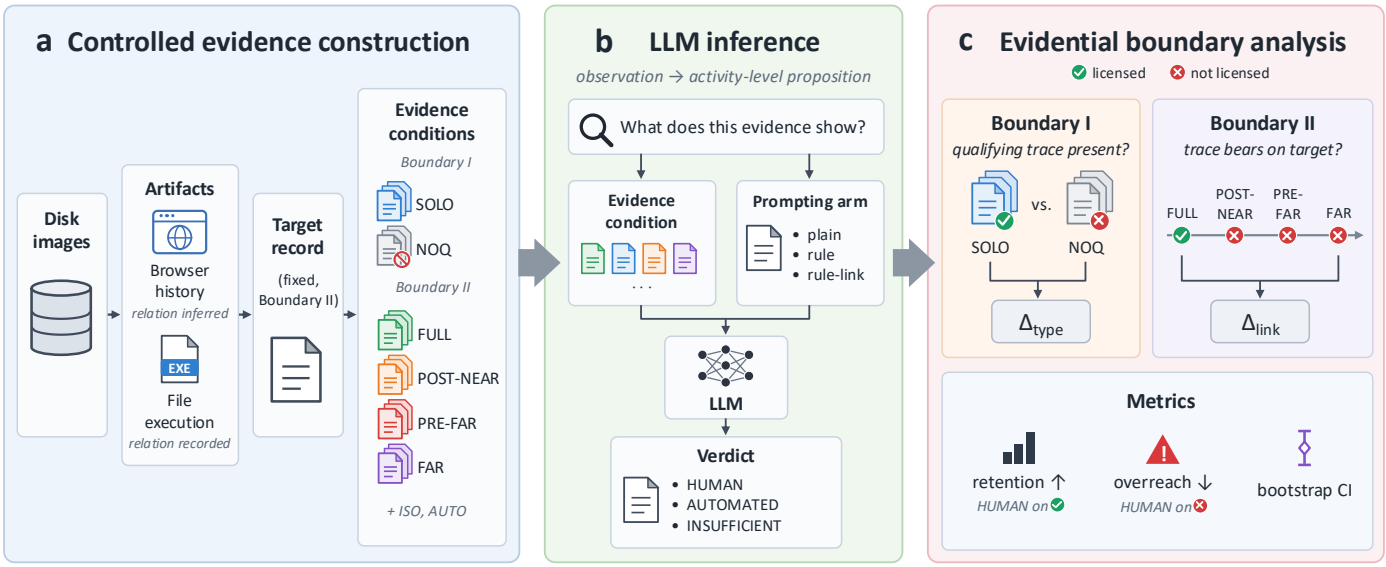


Figure 1: The protocol in three steps. (a) Conditions are built, not collected. From two public disk images, and around a target record whose annotation is fixed in advance, the eight conditions of Table 2 are carved. The four that carry Boundary II share that target and differ from FULL only in the qualifying record, while Boundary I compares two different records. The two artifact families differ in what that record gives: in browser history the relation to the target must be inferred, while the execution artifact records it. (b) Each condition and arm goes to each model, which returns one of three verdicts. (c) A tick marks a condition whose evidence licenses the HUMAN claim and a cross one that does not; the two contrasts take one conjunct of $L = K \wedge R$ each and are read beside retention and overreach. No measured quantity appears in this figure.

instrument, a deterministic oracle for trace type, is included because its failure is informative.

Our contributions are the following.

1. We define inferential overreach as a proposition-level failure that can occur without fabricated artifacts or incorrect citations, and give evidence conditions that measure whether a generator crosses an evidential boundary. Unsupported verdicts follow from what was deleted or swapped and need no per-item labelling; HUMAN target annotations come from the browser's visit graph and, where the corpus has one, its published ground truth, under a policy fixed in advance.
2. We separate evidence volume from evidential kind, and the presence of a qualifying trace from the relation it stands in to the record at issue. The second is tested with two decoys failing in opposite directions, so a separation surviving both is not reducible to proximity or precedence alone.
3. We give the reporting protocol this implies: retention beside overreach, because a low overreach rate is not by itself evidence of discrimination. On the main browser corpus the model-averaged fall in SOLO retention under a property-naming criterion exceeds the fall in overreach, in percentage points. Naming the relation recovers retention on FULL and yields contrasts at least as large on thirteen of the fifteen model–decoy comparisons in Table 5.

2. Background and related work

2.1. Forensic proposition levels

The recovered artifacts themselves are *observations*, and whether they can be mapped onto source-level propositions is a separate and contested question [7, 20] on which we take no position: the transition studied here runs from an observed trace to an activity-level proposition without passing through the source level, and the hierarchy matters because the error consists in making that crossing without meeting its evidential requirements.

The hierarchy [1] is the frame within which ENFSI requires evaluative opinions to be expressed [3], and its digital application is recent and thin. Casey [7] proposed standardising preliminary evaluative opinions for digital evidence; Sunde and Dror [8] analysed the human factors that shape digital interpretation. The most direct work is by the Netherlands Forensic Institute: Vink et al. formulate propositions for Trojan horse defence cases, distinguishing intentional and unintentional acts by the suspect, acts by another person, and acts of an automated process [4], and evaluate such findings with Bayesian networks [5]. A recent systematisation of likelihood-ratio practice for digital evidence [6] classifies existing work by proposition level but addresses statistical trace comparison; narrative conclusions produced by a generator fall outside its scope. All of this work is manual and addressed to human examiners. We borrow the human/automated distinction directly from [4]; our contribution is not the axis but its use as a constraint on generation.

2.2. LLM-assisted digital forensics, and evidential sufficiency

Recent work applies language models to forensic analysis tasks [9], timeline reconstruction [10], and the reasoning traces they produce [11]; a review notes that "only a few works explore multiple model architectures comparatively" [21]. Reliability there is usually framed as hallucination or unfaithful citation. Two systems constrain a model before it goes too far: HunterAgent [12] grounds each proposed causal link against surviving telemetry, and OpenSec [13] scores incident-response agents on an evidence-gated action rate. CAFE [14] verifies claims an examiner supplies instead of constraining what may be generated.

A parallel literature asks whether a model should answer or abstain given retrieved evidence [15, 16], whether an answer is entailed by its context [18, 19], and whether ablating cited evidence reveals reliance on parametric knowledge [17]. Outside QA, Peters and Chin-Yee [22] document generalisation beyond the evidence in scientific summarisation, using a binary contrast and not an ordered hierarchy. Across all of this work the decision is binary on one axis: it neither distinguishes an observation from a proposition about an act nor names the trace class that licenses the transition.

3. Method

3.1. Task and corpus

We use the NIST CFReDS Data Leakage Case [2], a scripted Windows scenario whose published answer key documents the actions performed. The material here is the Chrome history of the scripted user, read both as the artifact records an examiner would see and, for annotation only, as the visit graph the browser wrote alongside them.

Every question has the same form. Given a set of history records with one marked as the record under examination, does the evidence support treating that record as the user-intended endpoint of deliberate interactive navigation, rather than as an automatically generated intermediary or background browser navigation? The permitted answers are `HUMAN`, `AUTOMATED` and `INSUFFICIENT`. Scoring is exact match and no free text is judged.

`HUMAN` denotes evidence licensing an endpoint associated with interactive human navigation, as against an automated or background one. That is the construct measured throughout, and we do not take `HUMAN` as evidence of intention towards the exact address. The question above is the prompt verbatim, and its phrasing is stronger than the construct. Its wording is identical across every arm and condition, but its reading may interact with the criteria the arms add, so retention measures agreement with this licensing policy under this wording and not a verified loss of justified casework conclusions.

We rebuild the conditions on a second corpus: M57 Jean (NPS-2008-M57-JEAN, Digital Corpora [26]), a Firefox profile from 2008. Its `moz_historyvisits` table carries `from_visit` and `visit_type`, and the admission policy below transfers without change of substance to a visit at the end of a redirect chain reached directly from a search. The answer key does not transfer. M57 has no action-level ground truth, so the supported side there rests on the visit graph alone, with the relation criterion supplying the rest. The replication is reported separately (Section 4.4) and not pooled into the main results, except where stated.

A third set of conditions leaves browser history altogether. On the same M57 Jean image the prompt asks whether the record is an execution a person initiated, as against one started by a service, a scheduled task or another program, and two artifact classes stand where the two browser ones did. A Windows prefetch file records that a program ran whether a user, a service or the scheduler started it [24, 25]: the event, not the agent. A `UserAssist` `UEME_RUNPATH` value records a launch through the Explorer shell, with a run count, and is used here as evidence of a user-associated shell launch under its documented semantics [23, 24]. Ten executables appear in both. Because linkage here is identity rather than timing there is one decoy and not three, a qualifying record naming a different program. There is no answer key here either, so the supported side rests on the documented meaning of the artifact classes. `HUMAN` is defined at the level of executable identity: a matching `UserAssist` record supports a user-associated shell launch of that executable, not the attribution of this particular recorded execution to a person.

The two families are asymmetric by design. Here the relation is executable identity written into the record, making the verdict a function of what the model is shown. A device that asks whether a qualifying `UserAssist` record names the executable under examination reproduces all forty execution verdicts. One that asks only whether such a record is present reproduces thirty, failing on exactly the ten where the qualifying record names a different program. That makes the execution family the construct anchor. In the browser family the relation must be inferred across records; we specify it in advance as an episodic one and treat it as the harder setting, not as recovered provenance.

3.2. Targets and their annotation

A target is a Chrome *visit*, identified by visit id, not by URL: two records of the same address at different times are different events. A visit is admitted as a `HUMAN` target on two conditions. The visit graph must place it at the end of a navigation chain and not inside one, carrying `CHAIN_END` or `CHAIN_START` without a redirect qualifier. And the answer key must document user activity in that episode. Intermediate redirect hops are not admitted; they appear instead as the AUTO condition, whose verdict is `AUTOMATED`.

Applying that policy leaves six targets. Table 1 audits all eight candidates it considered, so that each annotation can be checked against the evidence that fixed it instead of taken on trust. All six admitted targets carry a `LINK` transition, which Chromium defines as a navigation reached by clicking a link [27]; most also carry redirect qualifiers and sit at the end of a chain whose first visit carries that transition. The policy reads that as an interactive endpoint, which is a reading of the recorded transition and not an observed click. One further can-

Table 1: Audit of target admission: every candidate the selection rule considered, with the evidence that fixed its annotation. END is CHAIN_END, START is CHAIN_START, CLI/SRV the redirect qualifiers, *key* whether the answer key documents user activity in the episode, and *offset* the FULL search's time from the target in seconds. Admission requires END, or START without a redirect qualifier, in a key-documented episode. The key holds for all eight candidates, so within these episodes the visit graph decided admission: the key constrains which episodes may contribute, not which visits inside them. E3T2 is the record quoted in Section 1.

target	visit	transition	key	offset	query group	target host	admitted
E2T1	36	LINK END,SRV	yes	−9	data leakage methods	sans.org	yes
E2T2	43	LINK END,CLI	yes	−21	information leakage case	emirates247.com	yes
E2T3	49	LINK END,CLI	yes	−6	information leakage case	mediapost.com	yes
E3T0	94	LINK END,CLI	yes	−9	apple icloud	apple.com	yes
E3T1	95	LINK START,END	yes	−19	apple icloud	apple.com	yes
E3T2	96	LINK START	yes	−25	apple icloud	icloud.com	yes
E2T0	35	LINK CLI	yes	n/a	data leakage methods	sans.org	no: mid-chain hop
E3T3	96	LINK START	yes	n/a	apple icloud	icloud.com	no: visit 96 above

Table 2: The conditions. The first five share a target and its three nearest plain neighbours and differ only in the qualifying interaction record, so the unsupported verdicts follow from what was swapped or removed. Offsets are the supplied search's time from the target in seconds, negative before it, and *n/a* marks conditions supplying no search. POST-NEAR is the closest available magnitude to FULL and breaks the order, PRE-FAR the reverse; no search in the corpus matches both.

condition	interaction record	n	offset (s)	verdict
FULL	its own earlier search	6	$[-25, -6]$	HUMAN
POST-NEAR	same session, later	6	$[+20, +46]$	INSUFFICIENT
PRE-FAR	same session, unrelated	5	$[-435, -83]$	INSUFFICIENT
FAR	a different session	6	$[-6764, +6325]$	INSUFFICIENT
NOQ	none; a fourth neighbour	6	n/a	INSUFFICIENT
ISO	none; the target alone	6	n/a	INSUFFICIENT
SOLO	documented search, alone	6	n/a	HUMAN
AUTO	a redirect hop, alone	8	n/a	AUTOMATED

didate was excluded as a mid-chain hop and one as a repeat of an already-admitted visit.

Every supported-side claim in this paper is a claim about the benchmark-defined licensing of Section 3.1, fixed by a policy set in advance. Whether it also tracks what the user did is a question the corpus cannot settle (Section 5.1).

The transition field bears directly on the question the models are asked. That is why they are not shown it: the field is what fixes the annotation, so supplying it would put the labelling rule into the input and turn the task into reading a field rather than weighing evidence. The model receives the record as a URL, a timestamp and a title. That is also the level at which browser history is commonly quoted in reports and summarised into a timeline, where rows from different artifact types are merged on time and the fields that relate one row to another are not carried across. Overreach here is measured on a representation poorer than a full `visits` row, and the result to read off it is not that the models are weak but that a pipeline which drops relation-bearing metadata leaves grounding with nothing to prevent a proposition-level error.

3.3. Evidence conditions

Table 2 lists them. For the observation-to-activity transitions studied here we operationalise licensing of proposition p by evidence set E as

$$L(p, E) = K(p, E) \wedge R(p, E),$$

K holding when E contains a trace of the kind p requires and R when that trace stands in the required relation to the record p is about. This is how the benchmark decides, not a general account of forensic evidential reasoning. Overreach is a model asserting p where $L = 0$. Each boundary targets one conjunct, and a device whose only variable is trace type computes K and has no term for R, so it cannot separate conditions differing in R alone. That follows from the decomposition and is not a finding about models.

Boundary I: volume against kind. SOLO is a single record that licenses the claim; NOQ is five that do not. The two are different records and not counterfactuals of one target, so we read $\Delta_{\text{type}} = P(\text{H} \mid \text{SOLO}) - P(\text{H} \mid \text{NOQ})$ as a diagnostic contrast and not as a causal effect.

Boundary II: presence against record relation. Here the conditions are counterfactuals of one target, whose annotation is fixed once by the admission policy of Section 3.2 and holds in all of them. Only R varies. FULL supplies the search that precedes the target by 6 to 25 seconds in the same scripted episode; each decoy keeps the target, its neighbours and the record count, and swaps that search for one the episode structure does not associate with the target. The provenance shows the target to be an interactive-navigation endpoint, not that this search produced it, so the relation we operationalise is episodic association, and *linkage* throughout means that benchmark-defined relation and not a record's causal provenance. The three decoys swap only the search:

POST-NEAR the same session at the closest available interval, but *after* the target, so it fails the benchmark's preceding-search criterion. Interval magnitude is as close as the corpus allows; temporal order is not.

PRE-FAR the nearest *preceding* search on a different query. Order is matched; interval is not, and we report the gap.

FAR a search from another session, as a stress test.

In two of the six targets the search selected for POST-NEAR repeats the query of the one it replaces, at a later time. The condition still fails the preceding-search criterion, but the decoy is weaker there because the evidence still shows the user issuing that query in the session.

Candidates were selected by a rule fixed before any model was run: among records satisfying a condition, take the one whose interval from the target is closest to the FULL interval. Where a target admits no candidate the cell is left absent, which is why PRE-FAR has five items.

Verdicts. On the unsupported side the verdict follows from what was removed or swapped and nothing is labelled by hand. On the supported side it cannot, because adding evidence does not make a claim true; there the verdict is the target annotation of Section 3.2 together with the relation criterion above, and SOLO uses only queries the answer key records verbatim.

Table 3 shows the record of Section 1 as the model receives it, condensed, under FULL and under PRE-FAR.

3.4. Methods compared

All arms see identical evidence and answer from the same three options. The criterion arms state what must be shown, not which artifact shows it, so that none of them restates the benchmark's assignment rule. Four arms carry the argument: PLAIN, a criterion naming a property the evidence must have, a criterion naming a relation it must stand in, and a deterministic type-only control. The others vary one feature of those at a time, so that the feature carrying an effect can be identified and not assumed.

PLAIN the question and the options.

SUFFICIENCY adds "if the evidence shown does not distinguish between the options, answer `INSUFFICIENT`".

GATE-GENERIC asks a content-free prerequisite first (whether the evidence supports a claim about what a person did) and puts the activity question only if it holds. This is decomposition without a criterion.

RULE names the criterion in one prompt: answer `HUMAN` only when the evidence affirmatively shows deliberate human interaction; evidence that a page was loaded is not, by itself, evidence that a person acted.

GATE asks that same criterion as a separate first question and puts the activity question only if it holds, returning `INSUFFICIENT` otherwise. Following ENFSI, a failed prerequisite does not restate the proposition at a lower level; it withholds a conclusion about the one at issue [3]. The activity question is then put without further constraint, making the arm a prerequisite-gated version of PLAIN.

RULE-LINK the criterion of RULE plus one clause: the interaction must bear on the record under examination, and an unrelated interaction elsewhere does not count. No timing, session or provenance cue is given. It runs on the Boundary II conditions only, where the qualifying search is always a record other than the one under examination and the clause has something to discriminate. On Boundary I it holds or fails trivially: SOLO's record under examination is itself the search, and NOQ and ISO hold no qualifying trace.

Two further arms are diagnostics. RULE-TYPE and GATE-TYPE repeat RULE and GATE with the trace class named: "a typed query, an explicit search, a submitted form", which is the benchmark's own assignment rule; they show what a method reaches when told exactly what to look for. *type-oracle* is a structural control and not a method: it answers `HUMAN` whenever a qualifying trace is present and `INSUFFICIENT` otherwise, deterministically and without calling a model. We report it to make the limit of Section 3.3 explicit.

3.5. Metrics and analysis

Write $\hat{y}$ for a sampled answer. The two contrasts are

$$\Delta_{\text{type}} = P(\hat{y}=\texttt{H} \mid \textsc{solo}) - P(\hat{y}=\texttt{H} \mid \textsc{noq}),$$
$$\Delta_{\text{link}} = P(\hat{y}=\texttt{H} \mid \textsc{full}) - P(\hat{y}=\texttt{H} \mid \textsc{decoy}),$$

the second computed against POST-NEAR and PRE-FAR separately and against FAR as a stress test.

A contrast cannot be read alone: answering `HUMAN` everywhere and answering `INSUFFICIENT` everywhere both score zero on both contrasts, and they are opposite failures. We therefore report each contrast beside the two quantities it is built from. *Retention* is the rate of `HUMAN` on the conditions whose evidence licenses the claim, SOLO for Boundary I and FULL for Boundary II. *Overreach* is the rate of `HUMAN` pooled over the five conditions whose evidence does not. A summary measure, *H/I balanced accuracy*, averages the rate of `HUMAN` where it is licensed with the rate of `INSUFFICIENT` where it is not, over those two verdicts because the question at issue is whether an activity-level `HUMAN` claim is licensed or withheld. AUTO is reported separately.

Each decoy contrast is taken on the targets admitting both FULL and that decoy, and the resampling preserves those within-target pairs, so a contrast measures the evidence swap and not a difference in which targets contribute. The unit of analysis is the Chrome visit. The conditions built on one target form a cluster, and intervals are bootstrapped by resampling the target clusters eligible for the contrast, not the cells within them; the five generations per cell are technical replicates and are not counted as independent items. The six main-corpus targets fall into three search-query groups, so resampling by target treats as independent some clusters that share an evidence record, while resampling by query leaves three clusters and a percentile interval with ten distinct values. We report target-level intervals for the main corpus and read them as optimistic. The replication corpus supplies eleven further query groups, and where the two are pooled we resample the fourteen and say so; those are the intervals the pooled claims of Sections 4.4 and 4.5 rest on. An answer from which no verdict can be parsed is kept in the denominator and counted as neither `HUMAN` nor `INSUFFICIENT`, since dropping it would reward a method for being unreadable.

The five generations also license a check on the draw itself. Vote index k across every cell is a complete single-sample replicate of the study, and we recompute each reported claim on all five. The contrasts and their signs, the retention collapse, the four-of-five separation under RULE-LINK and the replication's retention ordering hold on every replicate. Two statements do not,

Table 3: Model input and benchmark verdict for the record quoted in Section 1. The block above the rule is a condensed form of the model input; only the `[E2]` line differs between conditions, and its two forms are below with the label, which the model never sees. The plain neighbours here are scheme and slash variants of the target: E3T2 is the one admitted target whose nearest plain records are variants of itself, and for the other five they are distinct pages. Visit transition, `from_visit` and the answer key never reach the model.

`EVIDENCE:`
`[E1] 15:55:34 http://www.icloud.com/icloudcontrolpanel Title: iCloud`
`[E2]` *the one line that varies; see below*
`[E3]-[E5]` *the same address at 15:55:34 in its scheme and trailing-slash variants*
`The record [E1] is the page under examination.`
`QUESTION:` *the question and option set of Section 3.1, verbatim*

condition	the `[E2]` line	verdict (not shown)
FULL	`[E2] 15:55:09 google.com/webhp#q=apple+icloud`	`HUMAN`
PRE-FAR	`[E2] 15:48:19 google.com/webhp#q=google`	`INSUFFICIENT`

both about Qwen3.5 9B, whose estimates move by as much as 67 points between replicates, the widest spread of any model; they are marked where they appear.

3.6. Models

Five models, all current at the time of writing and drawn from four organisations. Two are open weights run locally: `gemma4:12b` and `qwen3.5:9b`. Three are hosted: `gpt-5.6-sol`, `claude-sonnet-5` and `gemini-3.8-flash`, accessed in September 2026. Five samples per cell at the providers' default sampling.

Decoding budgets are not matched across providers. Local models answer in twelve tokens; hosted models spend part of their budget on hidden reasoning and are given 1500 more, and the OpenAI model accepts only its default temperature.

Both corpora are run under the same arms and prompts: 2074 cells on the main corpus and 3685 on the replication. The design calls for 2075 on the main corpus; one cell, GEMMA4 under RULE-LINK on a single FAR condition, is absent from the run, and the paired aggregation below takes that contrast on the five targets that remain. Of the 28,795 sampled answers they produce, 26 carry no parseable verdict (0.09%). They are scored as above, and dropping them instead would move no reported quantity by more than 1.8 points.

Availability. Both corpora are public [2, 26]. Released with the paper are the condition-building scripts, the prompts verbatim, the raw model output behind every cell, and the script that recomputes every claim on each single-sample replicate, so that the conditions, the parsing and the stability of each number can be checked independently.

4. Results

The intervals of Section 3.5 are wide, and a model's position is evidence that a behaviour occurs rather than an estimate of where that model sits. On this corpus the unparseable answers of Section 3.6 number eight, all GEMMA4 under RULE. They move no rate of `HUMAN`, and so nothing in the tables; they move the balanced accuracy of Section 4.3 by 1.0 point.

The results are ordered as discovery, replication and pooled confirmation. Sections 4.1 to 4.3 establish the effects on the main corpus, where the answer key is available; Section 4.4 asks independently whether they recur on a second; and Section 4.5 pools the one condition both share, so that fourteen query groups carry the interval instead of three. We use the pooled query groups to summarise the cross-boundary profiles and to estimate the retention collapse; the near-decoy results are main-corpus estimates with a separate replication beside them. Section 4.6 then leaves browser history for file execution, where the relation is written into the record rather than inferred; it is the construct anchor for the decomposition, not an extension of it.

Table 4 carries the three axes for each model under the arms that span both boundaries.

4.1. What plain prompting does

Asked the activity question with no further instruction, every model answers `HUMAN` on a large share of the conditions whose evidence does not license it, from 39% for Sonnet 5 to 75% for Qwen3.5 9B. The `INSUFFICIENT` option was present in every prompt and was taken on 4% to 34% of those same conditions, so overreach is not an artifact of a missing option.

Boundary I: volume against trace kind. The models do not agree on which way this boundary runs. SOLO is one record that meets the criterion, NOQ five that do not; under PLAIN three separate them in the expected direction and two do not. Qwen3.5 9B is flat, answering `HUMAN` at about three quarters whatever it is holding. Sonnet 5 runs the other way, affirming interactive navigation more readily on five records of pages having loaded than on the single record of a typed query. No per-model interval excludes zero, and the finding is the split sign, not any one value.

Boundary II: presence against record relation. A model can track precedence, or session membership, without tracking the relation the proposition needs, and under PLAIN the decoys separate the models in different ways. Gemma4 12B clears POST-NEAR but not FAR, sensitive to precedence and indifferent to session membership, and Sonnet 5 is the reverse.

type-oracle scores $\Delta_{\text{link}} = 0$ against every decoy, as Section 3.3 says it must, and overreaches on 58.6% of the unsupported conditions.

Table 4: Retention, overreach and the two contrasts, in percent. Retention is %HUMAN where the evidence meets the criterion, given separately for the two such conditions; overreach is %HUMAN pooled over the five that do not. A contrast near zero is read against the retention columns beside it. Δ_{far} is the contrast against the session decoy on this corpus alone; Table 5 carries the near decoys, RULE-LINK, and the same FAR contrast pooled over both corpora, which is why its values differ. RULE-TYPE names the qualifying artifact and is a trace-explicit diagnostic, not a method; SUFFICIENCY and GATE-GENERIC are in Fig. 2 and GATE-TYPE in Section 4.2. *type-oracle* calls no model, and its row is identical for every model.

		retention		overreach	contrast, this corpus	
model	arm	SOLO	FULL	pooled	Δ_{type}	Δ_{far}
GPT-5.6-sol	plain	100.0	83.3	62.1	+26.7	+16.7
	rule	0.0	63.3	13.1	+0.0	+50.0
	gate	0.0	26.7	7.6	-3.3	+20.0
	rule-type	3.3	100.0	42.1	+3.3	+36.7
Sonnet 5	plain	13.3	66.7	39.3	-50.0	+43.3
	rule	10.0	53.3	9.0	+3.3	+53.3
	gate	0.0	66.7	29.7	-16.7	+50.0
	rule-type	83.3	100.0	9.7	+83.3	+100.0
Gemini 3.8 F	plain	96.7	66.7	57.2	+30.0	+3.3
	rule	0.0	36.7	0.0	+0.0	+36.7
	gate	0.0	46.7	4.1	-6.7	+43.3
	rule-type	23.3	73.3	6.9	+23.3	+73.3
Gemma4 12B	plain	100.0	83.3	69.7	+36.7	-6.7
	rule	0.0	0.0	0.0	+0.0	+0.0
	gate	0.0	20.0	6.2	+0.0	+16.7
	rule-type	83.3	56.7	3.4	+83.3	+56.7
Qwen3.5 9B	plain	73.3	73.3	75.2	-3.3	-6.7
	rule	56.7	40.0	49.0	+6.7	-13.3
	gate	36.7	40.0	44.8	+0.0	-13.3
	rule-type	80.0	43.3	21.4	+66.7	+20.0
any	*type-oracle*	100.0	100.0	58.6	+100.0	+0.0

4.2. Generic criteria suppress rather than discriminate

Naming the forensic criterion does not improve trace-type discrimination, and on several models it removes the claim altogether. On SOLO, where a single search record's type alone satisfies the criterion, HUMAN falls to zero for both GPT-5.6-sol and Gemma4 12B, a change of −100.0 points from PLAIN with a 95% CI of [−100.0, −100.0], and also to zero for Gemini 3.8 Flash, from a starting point of 96.7%, with GATE doing the same. After the criterion these three models never return HUMAN for a record that on its own meets it.

Δ_{type} under RULE is 0.0 for three of five models, and Table 4 shows why. Both SOLO and NOQ have gone to zero, which is not the same as telling them apart. If a criterion were teaching the distinction, retention would hold while overreach fell; the three panels of Figure 2 instead descend together. On GPT-5.6-sol, Gemini 3.8 Flash and Gemma4 12B retention on SOLO falls by more percentage points than overreach does, and on GPT-5.6-sol it reaches zero while 13.1% of unsupported conditions still draw a HUMAN answer.

Across the eight arms that run on SOLO, RULE-LINK being confined to Boundary II, mean retention there tracks one property and no other, which is whether the prerequisite names an observable that would satisfy it. The three arms that name none leave it at 7.3–13.3%, against 54.7–58.7% both for the two that name the qualifying artifact and for SUFFICIENCY, which requires nothing to be established at all. Averaged over the five models those three arms also give up more SOLO retention than they remove overreach, in percentage points, where naming the artifact gives up less than it removes. Forensic content does not separate the arms, and neither does whether the prerequisite is asked inline or apart.

What the criterion buys in exchange varies. Against the session decoy GPT-5.6-sol, Sonnet 5 and Gemini 3.8 Flash all raise Δ_{far} while keeping part of the supported claim (Table 4). Gemma4 12B's Δ_{far} of 0.0 is the degenerate case, its retention on FULL being zero as well and every parseable answer it gives in the study INSUFFICIENT. Qwen3.5 9B pays the cost without the gain. The near decoys that carry Boundary II are read in Section 4.3.

4.3. Relation-explicit criteria reduce the collapse

RULE-LINK differs from RULE in the one clause of Section 3.4, and Table 5 reads both arms against each decoy on both corpora. On the main corpus RULE-LINK separates FULL from POST-NEAR on four of five models but from PRE-FAR on two: the decoy preserving precedence is the one the criterion least often clears. Retention on FULL is not measurably reduced from PLAIN. Sonnet 5 rises, and although Gemma4 12B and Gemini 3.8 Flash fall, no model's interval excludes zero. Gemma4 12B, which collapses completely under RULE, recovers here.

Qwen3.5 9B is the exception throughout, though not a stable one: pooled over the five samples it returns HUMAN more often on POST-NEAR than on FULL, but on three of the five replicates the two are level rather than reversed.

The trace-class diagnostics bound the other end. Under RULE-TYPE, the trace-explicit diagnostic that names the qualifying artifact, the collapse is relieved on three of the five models and not on all. Retention on SOLO recovers to at least 80% on Sonnet 5, Gemma4 12B and Qwen3.5 9B and stays below a

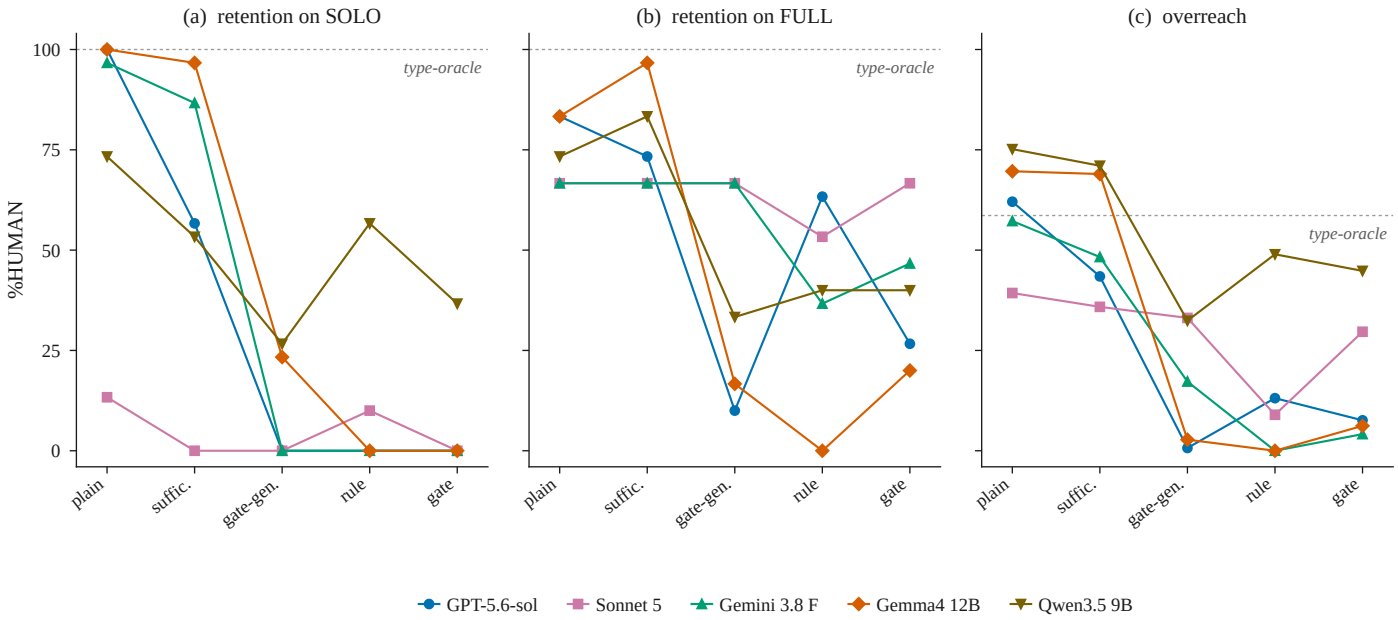


Figure 2: The three axes of Table 4 across the arms, on one scale: (a) retention on SOLO, (b) retention on FULL, (c) overreach. RULE-LINK runs on Boundary II only and is not plotted here; its results are in Table 5. A criterion that discriminated would hold (a) and (b) up while (c) fell; instead all three fall together, and on three models (a) falls furthest and to zero. The dashed line is *type-oracle*, which retains everything and still overreaches. Point estimates only.

quarter on the other two, so naming the artifact is enough for some models and not others, and where it works it works fully. Under RULE-TYPE the H/I balanced accuracy of Sonnet 5 and Gemma4 12B reaches 91.0 and 83.3, both above *type-oracle*'s 70.7. That figure is a limit on a type-only device and not a ceiling on models, since a model told the trace class can also weigh the relation.

RULE-TYPE can only be written by someone who already knows the answer, while RULE-LINK names a relation without restating the benchmark's assignment rule. Told what relation would settle the question, four of the five separate the near decoy and all five keep more of the supported claim.

4.4. Replication on a second corpus

The main qualitative patterns recur on a second corpus. The conditions rebuilt on M57 Jean give eleven targets over eleven distinct queries and four sessions, against six targets over three queries here. Two differences in that user's behaviour matter, neither of which enters the claims below: that user refines one topic across a session, so PRE-FAR is topically related to FULL for six of the nine targets that admit one, and the sessions are long enough that POST-NEAR still follows the target without its interval being matched to FULL.

Overreach under PLAIN is higher on the second corpus for every model, 76–92% against 39–75% here, and Boundary I discrimination nearly vanishes. With HUMAN returned on most of what the models hold, Δ_{type} falls within four points of zero for four of the five, against +23.6 for Gemma4 12B. The pooled estimates of Section 4.5 below are smaller than the main corpus alone for that reason, and we report them in place of the larger ones.

The collapse of Section 4.2 replicates, and on the models the main corpus could not show it for. Retention on SOLO under PLAIN is above 90% for all five there, where Sonnet 5 began near the floor on the main corpus and left nothing to suppress, and the criterion drives it down on every one. Pooling the fourteen query groups, every fall is clear of zero, the weakest of them −22.4 [−32.0, −11.8], where on the main corpus alone the intervals for Sonnet 5 and Qwen3.5 9B contained it.

Section 4.3 replicates on the retention side and not on the discrimination side. Retention on FULL under RULE-LINK is at or above its value under RULE on all five models, and Gemma4 12B, the one model whose retention collapses here too, roughly triples it. Overreach rises on all five as well: the relation clause moves the operating point towards asserting, repairing a model that had over-suppressed and costing one that had not. But RULE already leaves retention on FULL at 80% or above for four of the five here, against 63.3% at best on the main corpus, so there was little to repair. Separation from POST-NEAR improves on two of the five, and that is the limit of what the replication establishes about the relation criterion.

type-oracle reaches the same structural limit from different material, scoring $\Delta_{\text{link}} = 0$ against all three decoys there as here, on much the same share of the unsupported conditions.

4.5. Cross-boundary profiles under plain prompting

Figure 3 gives each model's two contrasts under PLAIN, both from the same prompt and pooled so that the fourteen query groups carry them. Exactly one model clears zero on each boundary (Gemma4 12B on the first, Sonnet 5 on the second) and none on both. Neither is Qwen3.5 9B, the model that behaves unlike the rest throughout, so the pattern does not rest on it. We did not test the model-by-boundary interaction, so this is a descriptive pattern and not a measured dissociation. What

Table 5: Δ_{link} against each decoy under the two criterion arms, with 95% intervals; bold marks an interval clear of zero. Retention is %HUMAN on FULL, the condition every contrast is taken from, so a contrast bought by suppressing the supported claim shows beside it. Retention is computed over all main-corpus FULL targets, while each decoy contrast is taken on the targets admitting both FULL and that decoy, with the resampling preserving those within-target pairs. The near decoys carry Boundary II. On the main corpus they are resampled over the target clusters eligible for each decoy; on the replication corpus they are not the same conditions (Section 4.4), so those two columns are point estimates over its ten and nine targets, given beside the main corpus rather than pooled with it. FAR is a search from another session in both corpora and is pooled over the fourteen query groups. PRE-FAR preserves precedence, which is why the criterion least often clears it.

		main corpus, by target			pooled	replication	
model	arm	retention	POST-NEAR	PRE-FAR	FAR	POST-NEAR	PRE-FAR
GPT-5.6-sol	RULE	63.3	**+43.3 [+23.3, +56.7]**	+28.0 [+0.0, +60.0]	**+57.6 [+45.9, +70.7]**	+32.0	+37.8
	RULE-LINK	76.7	**+53.3 [+26.7, +80.0]**	+32.0 [-8.0, +76.0]	**+65.9 [+53.8, +80.0]**	+32.0	+51.1
Sonnet 5	RULE	53.3	**+53.3 [+16.7, +86.7]**	+16.0 [-44.0, +76.0]	**+28.2 [+7.1, +45.7]**	+18.0	+13.3
	RULE-LINK	80.0	**+53.3 [+30.0, +80.0]**	**+36.0 [+8.0, +68.0]**	**+37.6 [+11.4, +59.0]**	+16.0	+8.9
Gemini 3.8 F	RULE	36.7	**+36.7 [+10.0, +66.7]**	**+44.0 [+16.0, +76.0]**	**+58.8 [+41.1, +78.7]**	+62.0	+60.0
	RULE-LINK	53.3	**+46.7 [+13.3, +80.0]**	**+48.0 [+20.0, +76.0]**	**+57.6 [+36.5, +80.0]**	+36.0	+57.8
Gemma4 12B	RULE	0.0	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]	+14.1 [+0.0, +33.8]	+14.0	+15.6
	RULE-LINK	50.0	**+50.0 [+16.7, +83.3]**	+20.0 [+0.0, +60.0]	**+56.2 [+34.7, +78.7]**	+58.0	+17.8
Qwen3.5 9B	RULE	40.0	-6.7 [-23.3, +10.0]	-16.0 [-40.0, +24.0]	-8.2 [-20.0, +3.3]	-4.0	-8.9
	RULE-LINK	70.0	-13.3 [-33.3, +6.7]	+0.0 [-16.0, +20.0]	+2.4 [-2.5, +7.8]	+2.0	+8.9

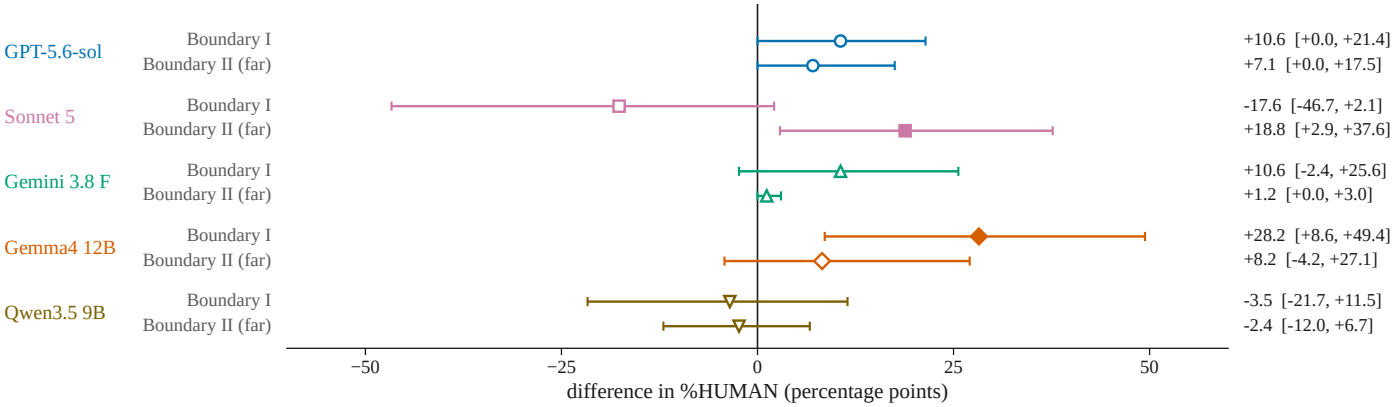


Figure 3: The two contrasts under PLAIN, pooled over both corpora, with 95% intervals over the fourteen query groups; a filled marker marks an interval clear of zero. Boundary II is the FAR-based contrast, the one condition the corpora share. Both values for a model come from one prompt, so each pair is read together, and the claim rests on the pattern across pairs rather than any one value.

it supports is narrower and enough: a positive result on one boundary does not by itself validate the other.

4.6. Explicit relations transfer across artifact families

Four arms were run on the execution conditions: PLAIN, RULE, RULE-LINK and *type-oracle*, over ten targets and five models. Two things carry over and one does not. *type-oracle* scores $\Delta_{\text{type}} = +100.0$ and separates FULL from the decoy by exactly 0.0: as in Section 3.1, a rule whose only variable is trace type has no term for which program the record names. The structural limit is therefore family-independent, holding wherever K is the only variable, not only on browser URLs. The retention collapse carries over too. Under RULE, retention on SOLO falls on all five models, by between 12 and 100 points, four of the five intervals clear of zero, and Gemma4 12B reaches 0.0% on SOLO here as it does on both browser corpora.

The empirical difficulty of Boundary II does not carry over. Under PLAIN four of the five models already answer `INSUFFICIENT` to the decoy on every item, separating it from FULL by between +68.0 and +100.0 points, with Qwen3.5 9B again the exception at −10.0, a value that is positive on two of the five single-sample replicates. Where the relation is written into the records, as when the `UserAssist` value names the executable, the models track it; where it must be inferred from timing, as in the browser conditions, they separate the decoys less often and never this cleanly. RULE-LINK adds little, the models already separating the decoy without it.

5. Discussion

Sufficiency is not a threshold, and caution is not discrimination. Instructing a model to abstain when evidence is insufficient adjusts how much is enough and moves both error directions together; K is not a quantity of that shape, since a set can be large and contain no qualifying trace, or minimal and contain one. A faithfulness measure does not catch the failure either: in SOLO and NOQ alike the context nowhere states that a person acted, and what separates them is a fact about how a typed query comes to exist. The arms separate on whether the prerequisite names something checkable, not on how much caution it expresses. A criterion naming only a property suppresses; one naming a relation to the record at issue keeps at

least as many HUMAN verdicts on FULL on every model in both browser corpora, though the gain in decoy discrimination does not replicate consistently. The suppressed models are not always unable to answer: named the artifact, three of the five keep it. ENFSI's guideline sets the duty this failure mode misses: the practitioner is to help the court by explaining the significance of the findings.

Type is necessary, not sufficient. Surviving both decoys is evidence against proximity and precedence as shortcuts, not a demonstration that the relation itself was used. The deterministic control makes the limit structural, and no improvement in recognising the right kind of trace can cross it. The type-only oracle also fails by construction in the execution family, where the relation is identity and no timing enters. Whether models fall over it turns on how the relation is recorded: read off an identity it is tracked, inferred from timing less often (Section 4.6).

Implications for LLM-assisted forensic systems. Three design rules follow from the measurements. *State prerequisites per proposition.* The same suppression reaches a verdict the criterion never mentions: three models identify redirect intermediaries as AUTOMATED on every item of the AUTO condition under PLAIN, and the criterion drops that as low as zero. A single report-level caution instruction is therefore the wrong unit. *Carry relation-bearing metadata into the representation the model is given.* Retrieving the artifact classes that are relevant in general is not enough, because what licenses an activity-level claim is the relation between a trace and the record under examination, and a timeline that merges rows from several artifact types on time drops exactly that. *Report supported-claim retention beside overreach.* A system that never overreaches because it never concludes anything passes an overreach metric and fails the examiner's task, and three of five models lose all SOLO retention under a plausible forensic instruction.

5.1. Limitations

Two artifact families, few targets in each. The execution family was chosen because its qualifying trace is unambiguous; families whose user-action artifact is itself contested, such as a registry key written by several mechanisms, may behave differently. Its conditions share an image with the browser replication, so they test transfer across artifact class and not across acquisition, and they differ in the class of the qualifying record as well as in how the relation is recorded, which localises the difficulty without isolating its cause. Both images are public teaching corpora with published walkthroughs, and we cannot rule out that a model has seen the scenario. Memorisation would bias towards HUMAN on the decoys, inflating overreach and not suppressing it, and the Boundary II contrasts are the less exposed for being built on a shared target.

The supported side is stipulated, not constructed. For the reason given in Section 3.3, the HUMAN annotation rests on the policy of Section 3.2. The relation separating FULL from the decoys is episodic in the same way: it rests on a search preceding the target by seconds within one scripted episode, and no admitted target's `from_visit` chain resolves to a typed query, stopping at a result redirector. A method could satisfy the criterion by tracking proximity, which is why the near decoys are read together and the claim limited to what surviving both establishes. No search in the corpus is preceding, unrelated and interval-matched, so neither decoy isolates the relation, and topical relatedness is uncontrolled.

What is not established here. RULE-LINK can only be tested on the Boundary II conditions (Section 3.4). We therefore know that naming a relation can improve discrimination without the collapse of Section 4.2 where the relation is non-trivial. We do not know whether it would prevent that collapse in general, and the conditions that would settle it need a supported claim resting on a trace both separate from the record at issue and of the kind Boundary I turns on, which this corpus lacks. The decoding asymmetry of Section 3.6 qualifies any comparison between a local and a hosted model. Answers come from a fixed option set, which holds the output format constant while measuring the judgement but leaves open whether overreach appears at the same rate in free narrative.

6. Conclusion

A language model asked to interpret a digital trace can cite the right artifact and still assert what that artifact does not license. We separate two requirements that evidential sufficiency runs together: that the evidence be of the kind an activity-level claim needs, and that it stand in a stated relation to the record being explained. Their contrasts pattern differently across models: one model's interval is clear of zero on the first boundary, a different model's on the second and none on both. A rule with access only to trace type cannot express the second however well it recognises the first.

Caution is not discrimination. On three of five models the forensic criterion drives retention on SOLO to zero, though two of them keep a HUMAN verdict on FULL, and the same criterion suppresses the positive finding of machine action. What separates a criterion that discriminates from one that suppresses is not how much caution it asks for but whether it names something an examiner could check. Carried to file execution, the structural limit and the collapse both reappear while the empirical failure does not: there the qualifying record names the program, and the relation is read rather than inferred. An overreach rate reported alone cannot tell a system that has learned what its evidence supports from one that lowers overreach by giving up supported conclusions.

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